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(54) **DENTAL IMPLANT HAVING TAPERED
THREADED SURFACE WITH
INSTALLATION ENHANCEMENT
FEATURES**

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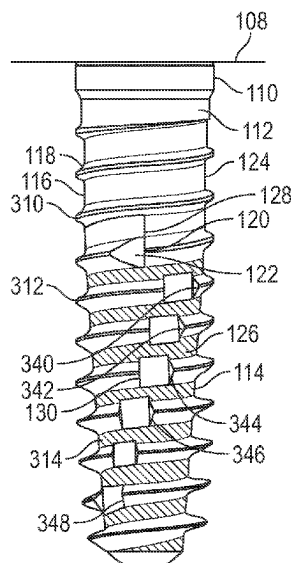
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(57) **ABSTRACT**

A dental implant with a specialized thread arrangement is disclosed. The dental implant includes a cylindrical upper section and a tapered lower section coupled to the cylindrical upper section. A helical thread is located on the exterior surface of the cylindrical upper section and the tapered lower section. A path is defined by the helical thread. The helical thread has a section on the exterior surface of the cylindrical body section and a section on the tapered body section. A cutting tooth is formed on the helical thread below where the thread section on the cylindrical body section transitions into the thread section on the tapered body section. The cutting tooth allows minimal torque to be applied to create stability of the implant by cutting into the dense part of the bone.

19 Claims, 2 Drawing Sheets



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- (60) Provisional application No. 62/203,645, filed on Aug. 11, 2015.

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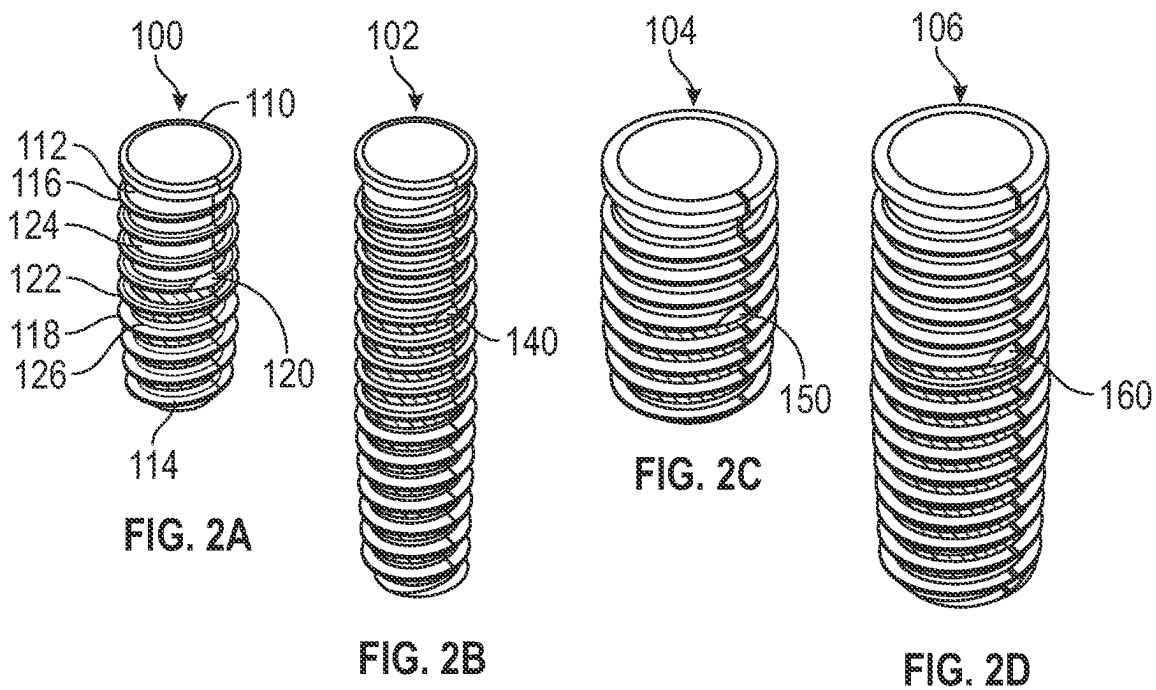
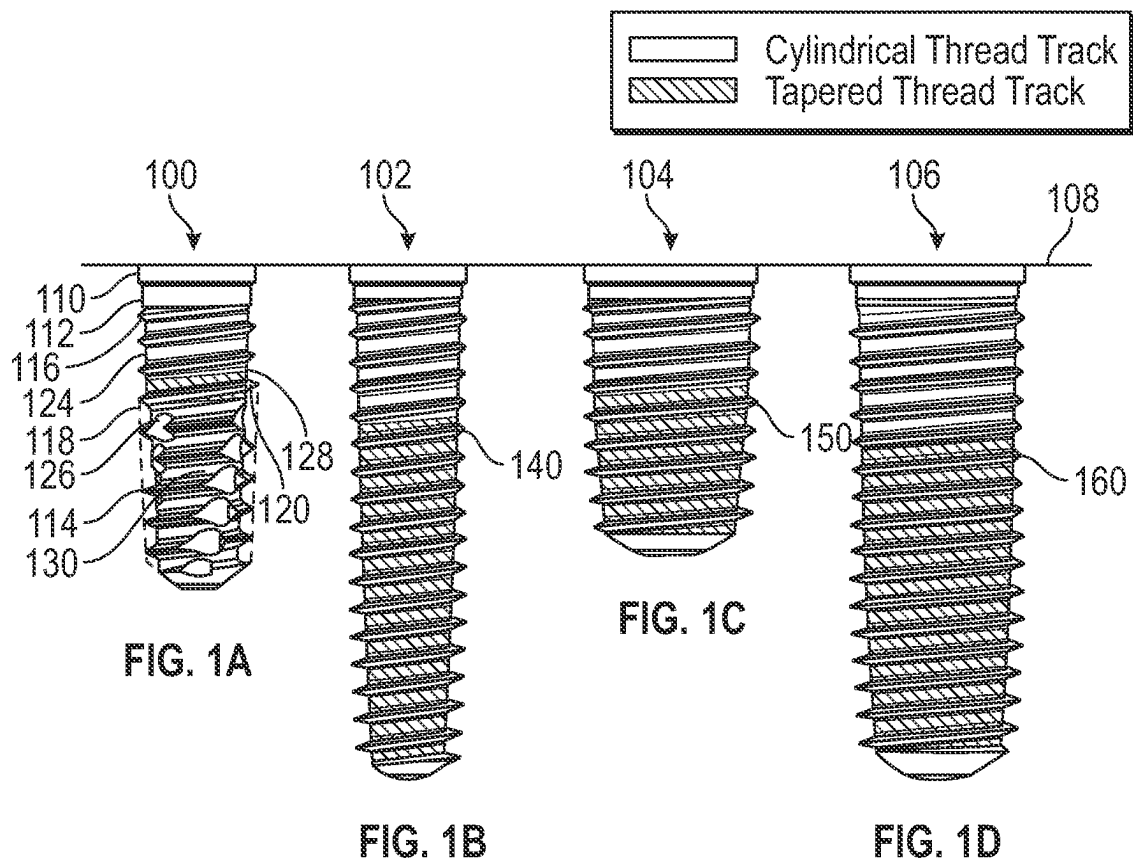
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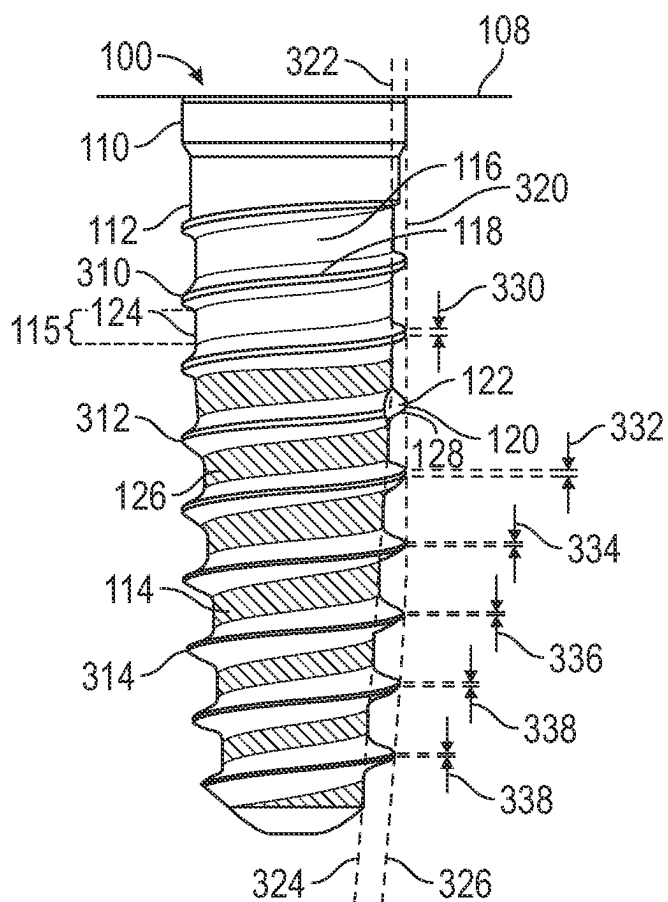


FIG. 3A

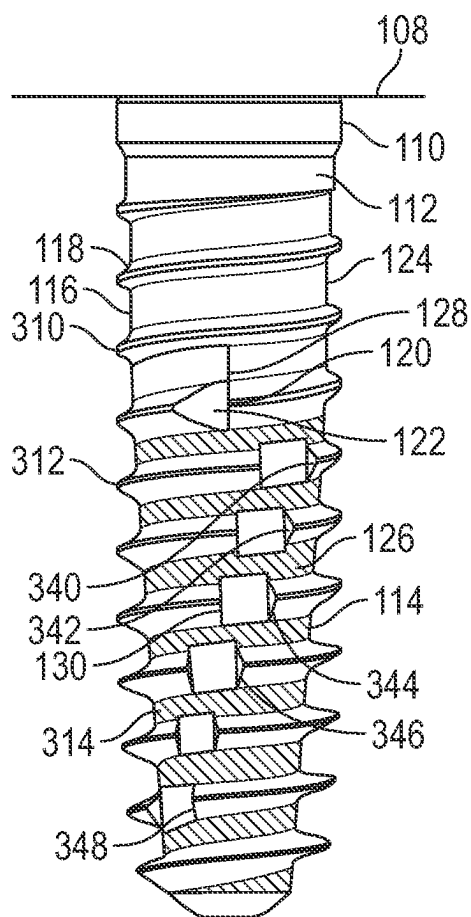


FIG. 3B

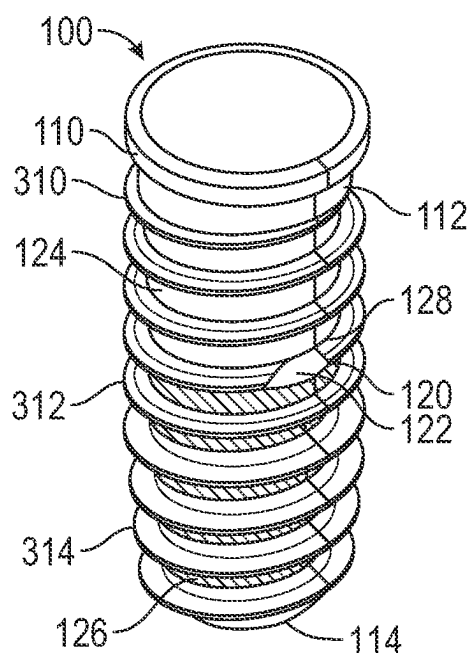


FIG. 3C

1

DENTAL IMPLANT HAVING TAPERED THREADED SURFACE WITH INSTALLATION ENHANCEMENT FEATURES

CROSS-REFERENCE TO RELATED APPLICATIONS

This U.S. Non-Provisional patent application is a Continuation of U.S. Non-Provisional patent application Ser. No. 15/233,655, filed Aug. 10, 2016, which claims the benefit of priority from U.S. Provisional Patent Application Ser. No. 62/203,645, filed on Aug. 11, 2015, the benefits of priority of which are claimed hereby, and the entire disclosures of which are hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

This disclosure relates to dental implants and more specifically to an implant with a tapered surface and a helical thread that allows efficient seating of the implant into bone.

BACKGROUND

A well-known procedure is the dental restoration of a partially or wholly edentulous patient with artificial dentition. Typically, a generally cylindrically shaped dental implant is seated into the bone of a patient's jaw in the initial stage of the restoration process. The dental implant includes a socket, e.g., a bore, which is accessible through the overlying or surrounding gum tissue for receiving and supporting one or more attachments or components which, in turn, are useful to fabricate and support prosthodontic restorations. The dental implant generally includes a threaded interior bore to receive a retaining screw for holding mating components thereupon. The seating of the dental implant in relation to the bone is critical as it provides stability for the restoration.

The exterior of an implant is designed to seat and stabilize the implant within the bone in order to anchor the eventual mating component. Generally, an osteotomy is created in the bone via a series of drills and/or taps and/or osteotomies in preparation for the seating of the dental implant. Certain implants include an exterior helical thread that creates a female thread in the sides of the osteotomy by compressing the bone when the implant is seated into the osteotomy. Such implants require relatively high torque to compress the bone via the threads and may not be optimal for the implant and/or the surgical instrument and/or the patient biology. In order to control torque for installation and the resultant stress at the bone-implant interface, it is desirable for the thread diameters to be close to the diameter of the osteotomy. Threads may be cut into the bone by the implant itself termed a "self-tapping" dental implant or by a tapping tool prior to the seating of the implant (or by both).

Known self-tapping dental implants have an exterior surface with a self-tapping thread that has cutting teeth that cut into the sides of the osteotomy to assist in seating the implant as torque is applied to the implant. Thus, the cutting teeth of the thread cut into the bone on the sides of the osteotomy to provide a female thread for seating of the trailing implant threads, and in turn the implant becomes stabilized within the osteotomy. The use of the cutting teeth decreases the required torque to seat the implant. However, the threading on certain dental implants may still cause compressive stresses on the bone as the cutting teeth are

2

typically located lower on the implant and therefore the threads on the upper part of the implant still use compression to cut through the bone to properly seat the implant (assuming the threads on the upper portion are at a diameter that is greater than the lower portion).

Certain implants have a cylindrical portion and are tapered along the length of the implant below the cylindrical portion such that the cylindrical portion of the implant has a greater diameter than the tapered bottom part of the implant inserted in the bone. Since the threads follow the exterior of the implant, the threads along the length of the implant are at different diameters due to the tapering geometry. Such tapered implants require additional care to avoid excessive cutting into the bone by the threads and increased compression as the larger diameter at the top of the implant enters the osteotomy. The placement of the self-tapping features on the threads on the tapered implant exterior is a challenge since the threads allow engagement with the dense bone, but may require additional torque on the implant that may not be optimal for the implant, the surgical instruments, or the patient biology. For example, if the cutting teeth are located above the transition between the cylindrical and tapered sections, some stability is sacrificed, since these teeth eliminate the thread portion but have no function, since the lower thread has already cut away the bone. If the cutting teeth are located below the transition between the cylindrical and tapered sections, the upper part of the thread still cuts via bone compression thereby increasing required torque.

Thus, there is a need for an implant having a self-tapping tapered threaded surface allowing reduced torque when the implant is seated in the bone. There is also a need for an implant with precision threads that allow engagement with dense bone in order to increase implant stability. There is also a need for a tapered implant that includes a specifically placed cutting tooth to allow for full stability for the non-tapered part of the implant yet minimizing bone compression along the same non-tapered part of the implant during placement. There is also a need for a tapered implant that allows the use of idealized osteotomy preparation tools such as drills and/or taps to optimize torque when seating the implant.

BRIEF SUMMARY

An example of the present disclosure is a dental implant including a cylindrical body section and a tapered body section coupled to the cylindrical body section. A helical thread is located on the exterior surfaces of the cylindrical body section and the tapered body section. The helical thread defines a path and has a first section on the cylindrical body section and a second section on the tapered body section. A plurality of self-tapping cutting teeth are located along the second section of the helical thread. The plurality of teeth include an uppermost cutting tooth formed on the helical thread where the first section on the cylindrical body section transitions into the second section on the tapered body section.

Another example disclosed is a self-tapping dental implant including a cylindrical body section and a tapered body section coupled to the cylindrical body section. A helical thread is located on the exterior surfaces of the cylindrical body section and the tapered body section. The helical thread defines a path located with a first section on the cylindrical body section and a second section on the tapered body section. An initial cutting tooth is formed on the helical thread where the first section on the cylindrical body section transitions into the second section on the

3

tapered body section. The cutting tooth is positioned to cut through bone when the dental implant is seated.

Another example is a method of seating an implant in bone. The implant includes a cylindrical section, a tapered section coupled to the cylindrical section, a helical thread member on the exterior surface of the cylindrical section and the tapered section, a path on the cylindrical section and the tapered section defined by the helical thread, and a cutting tooth formed on the helical thread where the thread transitions from the cylindrical section to the tapered section. An osteotomy having a diameter approximately the same as the minor diameter of the cylindrical section of the implant in the bone is created. The implant is seated in the osteotomy. Torque is applied to the implant so the cutting tooth contacts the bone to cut through a dense region of bone to create implant stability via the thread member on the cylindrical section.

The foregoing and additional aspects and implementations of the present disclosure will be apparent to those of ordinary skill in the art in view of the detailed description of various embodiments and/or aspects, which is made with reference to the drawings, a brief description of which is provided next.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing and other advantages of the present disclosure will become apparent upon reading the following detailed description and upon reference to the drawings.

FIG. 1A is a side view of a tapered implant with specialized threading and a specifically located initial cutting tooth that is part of the self-tapping feature;

FIG. 1B is a side view of a tapered implant with specialized threading and a specifically located initial cutting tooth that is part of the self-tapping feature;

FIG. 1C is a side view of a tapered implant with specialized threading and a specifically located initial cutting tooth that is part of the self-tapping feature;

FIG. 1D is a side view of a tapered implant with specialized threading and a specifically located initial cutting tooth that is part of the self-tapping feature;

FIG. 2A is a perspective view of the tapered implant in FIG. 1A with the specialized threading and the specifically located initial cutting tooth;

FIG. 2B is a perspective view of the tapered implant in FIG. 1B with the specialized threading and the specifically located initial cutting tooth;

FIG. 2C is a perspective view of the tapered implant of FIG. 2C with the specialized threading and the specifically located initial cutting tooth;

FIG. 2D is a perspective view of the tapered implant in FIG. 2D with the specialized threading and the specifically located initial cutting tooth;

FIG. 3A is a close-up side view of one of the dental implants in FIG. 1;

FIG. 3B is a rotated close-up side view of the dental implant in FIG. 3A; and

FIG. 3C is a close-up perspective view of the dental implant in FIG. 3A.

While the invention is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the invention is to cover

4

all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

FIGS. 1A-1D is a side view of different sized tapered dental implants **100**, **102**, **104**, and **106**. FIGS. 2A-2D is a perspective view of the dental implants **100**, **102**, **104**, and **106** shown in FIGS. 1A-1D. Each of the implants **100**, **102**, **104**, and **106** is a self-tapping implant designed to be seated in bone. The seating of the implants **100**, **102**, **104**, and **106** is shown relative to a horizontal line **108** that represents the crest of the bone. Each of the implants **100**, **102**, **104**, and **106** has a body having an upper cylindrical section and a lower tapered section. The exterior surfaces of the cylindrical and tapered sections define the minor diameter of the implant. A specialized helical thread is located on the exterior surfaces of the cylindrical and tapered sections. The specialized helical thread defines a major diameter of the implant as will be described below. Each of the implants **100**, **102**, **104**, and **106** may be fabricated from metal, metal alloys, ceramics, and combinations of these materials such as titanium (Ti) and titanium alloys.

The implants **100**, **102**, **104**, and **106** have similar shapes and features to facilitate stability of the implant in the bone and to minimize required torque to seat the implant in the bone. For example, the implant **100** includes a cylindrical body section **112** that is topped by a collar **110** and a tapered body section **114**. The tapered body section **114** is coupled to the cylindrical body section **112** opposite the collar **110**. A thread path or track **116** is formed between a helical thread **118** that runs continuously on the exterior surface of the cylindrical body section **112** and the tapered body section **114**. In this example, the implant **100** has a major diameter of 3.25 mm defined by the diameter of the thread **118** on the cylindrical body section and a length of 8.5 mm defined from the collar **110** to the tip of the tapered body section **114**. The thread **118** includes an initial cutting tooth **120** that cuts into the bone when the implant **100** is inserted into an osteotomy that is prepared in the bone. The cutting tooth **120** is defined by a vertical surface cut into the thread **118**. A triangular surface **122** is created on the thread **118** that is substantially perpendicular to the vertical surface of the cutting tooth **120**. The drills used to create the osteotomy have cutting diameters that are approximately the same as the minor diameters defined by the exterior surface of the cylindrical body section **112** and the tapered body section **114**.

The thread track **116** is divided between a cylindrical thread track **124** and a tapered thread track **126** at a transition point **128**. As may be seen in FIGS. 1A-1D and FIGS. 2A-2D, the cylindrical thread track **124** is defined by the thread **118** on the cylindrical body section **112**. The tapered thread track **126** is defined by the thread **118** on the tapered body section **114**. In this example, the cutting tooth **120** is located 3.53 mm from the top of the collar **110**. The cutting tooth **120** is thus adjacent to and lies half on the taper side of the junction of the minor diameters of the cylindrical and tapered body sections **112** and **114**. The depth of the cutting tooth **120** relative to the length of the implant **100** may be determined based on the values of the upper major diameter of the thread **118** on the cylindrical body section **112**, the bottom major diameter of the thread **118** at the bottom of the tapered body section **114**, the taper angle of the tapered body section **114**, and the overall length of the implant **100**.

5

The cutting tooth **120** thus is located on the thread **118** under the transition point **128** in the track **116**. The location of the transition point **128** and therefore the cutting tooth **120** depends on the diameter and length of the implant **100** in order to reduce drag from the thread **118** in relation to the osteotomy prepared in the bone. As shown in FIGS. 1A-1D, additional cutting teeth **130** are located on the thread **118** on the tapered body section **114**. The additional cutting teeth **130** provide cutting into the sides of the osteotomy to mate with the thread **118**.

The implant **102** has the same configuration as the implant **100** including a cutting tooth **140** at the transition point between the cylindrical track and the tapered track. The implant **102** has a major diameter of 3.25 mm but a length of 15 mm. The transition point on the implant **104** is thus 4.81 mm from the top of the collar as determined by the function described above. The location of the cutting tooth **140** at the transition point allows the tooth **140** to cut the bone for the thread above the tooth **140** to create stability in the dense part of the bone while minimizing required torque. The additional cutting teeth of the implant **102** are omitted for clarity in FIGS. 1A-1D and FIGS. 2A-2D.

The implant **104** has the same configuration as the implant **100** including a cutting tooth **150** at the transition point between the cylindrical track and the tapered track. The implant **104** has a major diameter of 5.0 mm and a length of 8.5 mm. The transition point on the implant **104** is thus 4.03 mm from the top of the collar as determined by the function described above. The location of the cutting tooth **150** at the transition point allows the tooth **150** to cut the bone for the thread above the tooth **150** to create stability in the dense part of the bone while minimizing required torque. The additional cutting teeth of the implant **104** are omitted for clarity in FIGS. 1A-1D and FIGS. 2A-2D.

The implant **106** has the same configuration as the implant **100** including a cutting tooth **160** at the transition point between the cylindrical track and the tapered track. The implant **106** has a major diameter of 5.0 mm and a length of 15 mm. The transition point on the implant **104** is thus 5.49 mm from the top of the collar as determined by the function described above. The location of the cutting tooth **160** at the transition point allows the tooth **160** to cut the bone for the thread above the tooth **160** to create stability in the dense part of the bone while minimizing required torque. The additional cutting teeth of the implant **106** are omitted for clarity in FIGS. 1A-1D and FIGS. 2A-2D.

The dimensions of length and major diameter for the implants described above are by example only. Of course implants having other lengths and major diameters may incorporate the cutting tooth location and thread design described herein. As explained above, each tapered implant such as the example implants **100**, **102**, **104**, and **106**, has a theoretical singular position at which the initial cutting tooth may be located in order to eliminate drag from the trailing threads in the dense part of the bone corresponding to the cylindrical body section of the implant. This selection thereby increases efficiency of seating the implant by minimizing required torque while maintaining stability of the seated implant via the self-tapped threads.

As may be seen by the other implants **102**, **104**, and **106**, regardless of the implant diameter or length, the first full-form tooth profile opportunity along the length of an example tapered implants **102**, **104**, and **106** is always found adjacent to (and on the taper side of) the junction of the minor diameters of the cylindrical and tapered body sections

6

of the respective implants. The succeeding cutting teeth such as the cutting teeth **130** are located on the thread **118** under the initial cutting tooth **120**.

FIGS. 3A and 3B are rotated close-up side views of the implant **100** in FIGS. 1A-1D. FIG. 3C is a close-up perspective view of the implant **100** in FIGS. 1A-1D. Like elements in FIGS. 3A-3C are assigned the same reference numbers as those in FIGS. 1A-1D and FIGS. 2A-2D. As may be seen in FIGS. 3A-3C, the thread **118** includes a cylindrical thread section **310**, an upper tapered thread section **312** and a lower tapered thread section **314**. The thread **118** has a roughly triangular shaped cross-section from the respective exterior surface of the implant body. The triangular shaped cross-section of the thread **118** defines the cutting tooth **120** as shown in FIGS. 3A and 3B. The additional cutting teeth **130** are formed in the thread **118** at locations below the initial cutting tooth **120** such as cutting teeth **340**, **342**, **344**, **346**, and **348** as shown in FIG. 3B. In this example, the cutting teeth **130** are located in relation to each rotation of the helical thread **118** to provide self-tapping in the wall of the osteotomy to seat the tapered body section **114** via the upper and lower tapered thread sections **312** and **314**. The additional cutting teeth **130** are not shown in FIGS. 3A and 3C for clarity.

Each of the thread sections **310**, **312**, and **314** has a major diameter that is the exterior surface of the thread and a minor diameter that is the location where the thread section joins the implant body. The cylindrical thread section **310** extends from the exterior surface of the cylindrical body section **112**. Thus, the minor diameter of the upper tapered thread section **312** is the same as the minor diameter of the cylindrical body section **112**. The tapered thread sections **312** and **314** extend from the exterior surface of the tapered body section **114**. The transition point **128** indicates the junction of a minor diameter of the cylindrical thread section **310** and the minor diameter of the upper tapered thread section **312**. In this example, the major diameter of the thread sections **310** and **312** are the same, as shown by a line **320** representing the major diameter of the thread sections **310** and **312**. The minor diameter of the cylindrical thread section **310** is represented by a line **322** that is also the minor diameter of the cylindrical body section **112**.

As explained above, the minor diameter of the cylindrical body section **112** is constant. However the minor diameter of the tapered body section **114** changes based on the taper angle as shown by a line **324**. The major diameters of the threads of the lower tapered thread section **314** are also tapered at approximately the taper angle represented by a line **326**.

As may be seen in FIGS. 3B and 3C, the location of the cutting tooth **120** coincides with the precise point on the thread **118** where the cylindrical thread track **124** meets the tapered thread track **126**. The crest width of the thread **118** in the cylindrical thread section **310** is constant as represented by a bracket **330**. Along the helical thread **118** below this point, the thread profile of the thread **118** as measured by the thread crest width decreases as a function of the thread profile. The increasing distance relationship between the major diameter and the minor diameter defined by the taper angle represented by the line **324** of the upper tapered thread section **312** causes this reduction in thread crest width shown in FIG. 3A. Since the minor diameter gets progressively smaller due to the taper of the exterior surface of the tapered body section **114** while the major diameter remains constant in the upper tapered thread section **312**, the thread crest continues to reduce in width to the point where the major diameter also tapers at the lower tapered thread

section 314. The reduction in thread crest may be seen by brackets 332, 334, and 336 that are successively smaller in width based on the location on the thread section 312. Once the major and minor diameters of the thread 118 both taper at the same rate in the lower tapered thread section 314, the thread crest remains constant as represented by brackets 338. This may be seen by the angles of the tapering of the minor and major diameters represented by the lines 324 and 326 in FIG. 3A. Due to the secondary threading cut that passes along the entire length of the implant 100 to thin out the straight thread, the whole region below the convergence point at the transition point 128 where the cutting tooth 120 is located is characterized by complex thread profile geometries of varying widths along the length of the tapered body section 114, while the root width 115 of the thread 118 remains constant as a function of the threading tool shape. The complex geometry of the thread sections 312 and 314 create a wedge effect with the sides of the osteotomy corresponding to the tapered body section 114 of the implant to generate stability for the seating of the implant 100.

The major and minor diameters remain constant above the transition point 128 and hence the thread profile of the cylindrical thread section 310 (as expressed by a constant crest width) also remains constant. The single tooth 120 at the transition point 128, therefore, cuts the full profile of thread 118 into the wall of the bone corresponding to the cylindrical body section 112 for the remainder of the cylindrical thread section 310 to follow until full seating of the implant 100 occurs in the osteotomy. This virtually eliminates any wedging of the full-form thread 118 of the cylindrical body section 112 into the wall of the cortical bone.

The osteotomy location for the implant 100 is selected in order to increase bone to implant contact to increase implant stability but also minimize the torque necessary to seat the implant 100. To increase stability of the implant, appropriate drills may be selected to create the osteotomy to seat the implant 100. Typically a starter drill is used to locate where the implant 100 will be placed. A second drill with a cutting diameter smaller than the final drill shape is used to excavate the site of the bulk of the bone to be removed. A final drill has a cutting diameter and shape that matches the minor diameter of the cylindrical body section 112. The location of the cutting tooth 120 at the point where the cylindrical thread track 124 meets the tapered thread track 126 increases placement efficiency without detracting from the implant stability. Drills of differing cutting diameters may be used to create osteotomies appropriate for the different dimensioned implants 102, 104, and 106 in FIGS. 1A-1D or implants of other dimensions.

Sometimes, the bone surrounding the osteotomy is so dense that the torque required to drive the implant cutting features overcomes the strength of the driving features of the implant, leading to damage and an inability to fully place the implant. In order to reduce this torque requirement, the site may be pre-tapped with a bone tap designed to remove the majority of the bone material in the internal thread track, leaving behind a small amount of bone for the implant itself to remove thereby requiring a greatly reduced amount of input torque. Thus, optimal stability may be achieved by using load sharing taps that create slightly undersized threads (vs. line to line with the implant macro-geometry) so as to make the implant 100 perform some work in cutting into the dense bone. While dense bone offers the greatest potential for high primary stability, known implants do not take full advantage of the dense bone for seating the implant because the osteotomies produced by the corresponding drill

for use in dense bone are oftentimes designed to remove a greater amount of bone, leaving a significantly larger diameter than the minor diameter of the implant. While this reduces the required torque to fully seat the implant, it also reduces significantly the initial bone to implant contact, which is so desirable for primary stability and optimal osseointegration. The disclosed implant 100 allows the creation of a tap to reduce the extremely high torques that would otherwise occur, but not so much that high stability that may be created in dense bone is sacrificed. The load sharing taps created for the implant 100 provide for a high primary stability and high efficiency based on the thread 118 and elimination of unnecessary cutting of dense bone by the implant requiring excessive torque.

The implant 100 also allows for shorter collar length of the collar 110 relative to known implants. The collar 110 by itself does nothing to contribute to the stability because it is not anchored into the bone as it does not have threads. The starting location of the thread 118 on the cylindrical body section 112 relative to the crest 108 in FIGS. 1A-1D is relatively high. Thus, the thread 118 at the cylindrical body section 112 may contact the best quality dense bone for stability of the implant 100. By moving the threads 118 up on the length of the implant 100, the threads 118 are able to bite into the cortical bone in contrast to cancellous or mushy bone, and this in turn increases the stability of the seated implant 100.

While the description above applies specifically to individual teeth cut into and along the helical length of the thread, there exists no restriction to the essence of the disclosed examples applying to other forms of tooth creation, such that the origin point of multiple teeth formed by a single cut into the length of the implant would lie at the same place as the description below is for a solitary tooth location, and, in fact, may even apply to a minimum of only one tooth should the implant be so designed. In such iterations, it is understood that the resultant shape of the surface 122 will be a function of the method of tooth creation and hence may differ somewhat from the triangular shape in the embodiments previously described.

While particular implementations and applications of the present disclosure have been illustrated and described, it is to be understood that the present disclosure is not limited to the precise construction and compositions disclosed herein and that various modifications, changes, and variations can be apparent from the foregoing descriptions without departing from the spirit and scope of the invention as defined in the appended claims.

What is claimed is:

1. A dental implant comprising:

- a cylindrical body section having a cylindrical body diameter that is at least substantially constant, the cylindrical body section being distal of and abutting against a collar of the dental implant;
- a tapered body section coupled to the cylindrical body section at a transition point, the tapered body section including a tapered body diameter which decreases from the transition point to a tip of the dental implant;
- a helical thread on exterior surfaces of the cylindrical body section and the tapered body section, the helical thread having a first section on the cylindrical body section that transitions uninterrupted into a second section on the tapered body section, wherein the helical thread defines a non-intersected path of at least one full rotation about the dental implant between adjacent thread peaks in the first section and in at least a portion of the second section; and

9

at least one cutting tooth positioned along the second section of the helical thread,

wherein the at least one cutting tooth is formed within a portion of a thread peak in the second section of the helical thread defined between two adjacent portions of the non-intersected path,

wherein the at least one cutting tooth is located at a position along the second section of the helical thread to require a torque to seat the dental implant, and

wherein the torque is dependent on a total length of the dental implant and the cylindrical body diameter of the cylindrical body section.

2. The dental implant of claim 1, wherein the at least one cutting tooth is an uppermost cutting tooth and is positioned on the helical thread under the transition point between the cylindrical body section and the tapered body section.

3. The dental implant of claim 1, wherein the at least one cutting tooth is self-tapping, wherein the at least one cutting tooth is defined by a vertical surface cut into the helical thread substantially parallel to a transverse axis through the dental implant, and wherein a triangular surface is formed on the helical thread that is substantially perpendicular to the vertical surface of the at least one cutting tooth and the transverse axis through the dental implant.

4. The dental implant of claim 1, wherein a root width of the non-intersected path is constant at least along a length of the first section of the helical thread on the cylindrical body section, and wherein the first section of the helical thread on the cylindrical body section includes a thread profile with a constant crest width defined by a major diameter and a minor diameter of the first section.

5. The dental implant of claim 4, wherein the second section of the helical thread on the tapered body section includes an upper tapered section and a lower tapered section.

6. The dental implant of claim 5, wherein the major diameter of the helical thread in the cylindrical body section and a major diameter of the upper tapered section is constant.

7. The dental implant of claim 6, wherein a thread profile of the upper tapered section is reduced as a distance increases from the transition point, wherein the thread profile is defined by a reduction in a thread crest width of the upper tapered section.

8. The dental implant of claim 7, wherein the reduction in the thread crest width is generated by an increasing distance between the major diameter and the minor diameter of the upper tapered section, as defined by a taper angle of the upper tapered section, where the minor diameter is reduced and the major diameter is constant as the distance increases from the transition point.

9. The dental implant of claim 5, wherein a thread crest width of the lower tapered section is defined by a minor diameter and a major diameter of the lower tapered section, wherein the thread crest width of the lower tapered section is constant as the minor diameter and the major diameter of the lower tapered section taper at a same rate.

10. The dental implant of claim 1, wherein the dental implant is fabricated from titanium or titanium alloy.

11. A method of seating a dental implant in bone, comprising:

creating an osteotomy for the dental implant in the bone, wherein the dental implant comprises:

a cylindrical body section having a cylindrical body diameter that is at least substantially constant, the cylindrical body section being distal of and abutting against a collar of the dental implant;

10

a tapered body section coupled to the cylindrical body section at a transition point, the tapered body section including a tapered body diameter which decreases from the transition point to a tip of the dental implant;

a helical thread on exterior surfaces of the cylindrical body section and the tapered body section, the helical thread having a first section on the cylindrical body section that is continuous with a second section on the tapered body section, wherein the helical thread defines a continuous path of at least one full rotation about the dental implant between adjacent thread peaks in the first section and in at least a portion of the second section; and

at least one cutting tooth positioned along the second section of the helical thread,

wherein the at least one cutting tooth is formed within a portion of a thread peak between two adjacent portions of the continuous path, wherein the portion of the thread peak is defined as less than one full rotation of the helical thread about a circumference of the dental implant,

wherein the at least one cutting tooth is located at a position along the second section of the helical thread to require a torque to seat the dental implant, and

wherein the torque is dependent on a total length of the dental implant and the cylindrical body diameter of the cylindrical body section,

wherein the osteotomy has a same diameter as a minor diameter of the cylindrical body section;

seating the dental implant in the osteotomy; and applying the torque to the dental implant so that the at least one cutting tooth contacts the bone to cut through a dense region of bone to create implant stability via the helical thread on the cylindrical body section.

12. The method of claim 11, wherein the at least one cutting tooth is an uppermost cutting tooth and is positioned on the helical thread under the transition point between the cylindrical body section and the tapered body section.

13. The method of claim 11, wherein the at least one cutting tooth is self-tapping, wherein the at least one cutting tooth is defined by a vertical surface cut into the helical thread substantially parallel to a transverse axis through the dental implant, and wherein a triangular surface is formed on the helical thread that is substantially perpendicular to the vertical surface of the at least one cutting tooth and the transverse axis through the dental implant.

14. The method of claim 11, wherein a root width of the continuous path is constant at least along a length of the first section of the helical thread on the cylindrical body section, and wherein the first section of the helical thread on the cylindrical body section includes a thread profile with a constant crest width defined by a major diameter and a minor diameter of the first section.

15. The method of claim 14, wherein the second section of the helical thread on the tapered body section includes an upper tapered section and a lower tapered section.

16. The method of claim 15, wherein the major diameter of the helical thread in the cylindrical body section and a major diameter of the upper tapered section is constant, wherein a thread profile of the upper tapered section is reduced as a distance increases from the transition point, wherein the thread profile is defined by a reduction in a thread crest width of the upper tapered section, and wherein the reduction in the thread crest width is generated by an increasing distance between the major diameter and the

11

minor diameter of the upper tapered section, as defined by a taper angle of the upper tapered section, where the minor diameter is reduced and the major diameter is constant as the distance increases from the transition point.

17. The method of claim 15, wherein a thread crest width 5 of the lower tapered section is defined by a minor diameter and a major diameter of the lower tapered section, wherein the thread crest width of the lower tapered section is constant as the minor diameter and the major diameter of the lower tapered section taper at a same rate. 10

18. The method of claim 11, wherein the torque is selected to reduce an amount of force required to seat the dental implant in the bone.

19. The method of claim 11, further comprising pre-tapping the osteotomy with a bone tap. 15

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12